Speaker Information

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Talk Information

Inferring interactions across multi-regional neural population under motor control

Dexterous movement depends on coordinated activity across multiple cortical regions, yet these interactions are often analyzed only in pairs. Using simultaneous large-scale recordings during mouse reach-to-grasp behavior, we combined latent-factor modeling, reference-frame alignment, and empirical dynamic modeling to infer cross-area interactions among motor cortex (M1), premotor cortex (M2), and primary somatosensory cortex (S1). The analysis revealed stage-specific, time-varying interaction motifs spanning movement preparation, reach onset, and grasp. Inter-areal coupling also differed between successful and failed trials, consistent with divergences in neural state-space trajectories. Removing cross-area interactions reduced the accuracy of reconstructing paw trajectories, indicating that distributed cortical communication is essential for precise motor output.

Research Overview (for collaboration discussion)

[Current Research Projects] STN population dynamics and optimization of deep-brain stimulation in Parkinsonian models; chronic CMOS-mated microwire arrays for multi-regional recording and targeted stimulation; single-trial intercortical dynamics during dexterous forelimb movement; CASTLE for unsupervised behavioral classification; layer-specific astrocyte specialization in M1.

[Techniques] In vivo two-photon calcium imaging and longitudinal spine imaging; optogenetics, DBS/oDBS, and Parkinsonian mouse models; chronic high-density electrophysiology with microwire-CMOS interfaces; behavior tracking and machine learning; LFADS, manifold alignment, and empirical dynamic modeling; scRNA-seq, CRISPR/Cas9, and super-resolution microscopy.

[Aspects or techniques of interest for future collaboration] Tissue clearing and whole-brain mapping; advanced optical and electrical microscopy; spatial transcriptomics; computational interpretation of large-scale neural datasets; neural interfaces and precision neuromodulation.

[Short Bio] Yu-Wei Wu received his Ph.D. in Neurology & Neuroscience from University College London and trained at RIKEN Brain Science Institute (BSI) and Stanford University. He is Assistant Research Fellow at Academia Sinica, where his lab integrates in vivo imaging, electrophysiology, neuroengineering, and computational modeling to study movement control, Parkinson's disease, astrocyte biology, and next-generation brain-computer interfaces.